

IN VITRO STUDIES OF CELLULAR AND MOLECULAR DEVELOPMENTAL TOXICITY OF ADJUVANTS, HERBICIDES, AND FUNGICIDES COMMONLY USED IN RED RIVER VALLEY, MINNESOTA

N. Lin, V. F. Garry

Environmental Medicine and Pathology, University of Minnesota,
Minneapolis, Minnesota, USA

Recent epidemiologic studies showed increased frequency of birth defects in pesticide applicators and general population of the Red River Valley, Minnesota. These studies further indicated that this crop growing area used more chlorophenoxy herbicides and fungicides than elsewhere in Minnesota. Based on frequency of use and known biology, certain herbicides, pesticide additives, fungicides, and mycotoxins are suspect agents. To define whether these agents affect developmental endpoints in vitro, 16 selected agrochemicals were examined using the MCF-7 breast cancer cell line. In the flow cytometric assay, cell proliferation in this estrogen-responsive cell line indicates xenobiotic-mediated estrogenic effects. Cell viability, morphology, ploidy, and apoptosis were incorporated in this assay. Data showed that the adjuvants X-77 and Activate Plus induced significant cell proliferation at 0.1 and 1 µg/ml. The commercial-grade herbicides 2,4-D LV4 and 2,4-D amine induced cell proliferation at 1 and 10 µg/ml. The reagent-grade 2,4-D products failed to induce proliferation over the same concentration range, suggesting that other ingredients in the commercial products, presumably adjuvants, could be a factor in these results. The fungicides triphenyltin and mancozeb induced apoptosis at concentrations of 4.1 µg/ml (10^{-5} M) and 50 µg/ml, respectively. Triphenyltin also induced aneuploidy (G2/M arrest) at 0.41 µg/ml (10^{-6} M). Data provide a mechanistic step to understanding human reproductive and developmental effects in populations exposed to these agrochemicals, and initiative to focusing limited resources for future in vivo animal developmental toxicity studies.

A number of pesticides have been found to induce concentration related developmental toxicity in animals (Murakami & Fukami, 1986; Wang, 1988; Roeleveld et al., 1990). Similarly, a growing body of epidemiological studies suggests that pesticide exposure may be associated with increased risk of birth defects in exposed human populations (Mattison et al., 1990; Czeizel et al., 1993; Garry et al., 1996; Kristensen et al., 1997; Poynor et al., 1997). In continuing studies conducted by our laboratory, it was found that the children born to pesticide applicators exposed to high levels of

Received 25 November 1999; sent for revision 5 January 2000; accepted 6 March 2000.

We thank Drs. B. Leung, A. Lyubimov, and W. Jaszcz for helpful discussions, L. Erickson and Dr. J. Coad for help on microscopic slide preparation and photography, and J. Prybil and B. Burroughs for technical assistance. This work is supported partly by National Institutes of Health (NIH) grant 5R01-ES08161.

Address correspondence to V. F. Garry, Environmental Medicine and Pathology, University of Minnesota, 421 29th Avenue SE, Minneapolis, MN 55414, USA. E-mail: garry001@maroon.te.umn.edu

chlorophenoxy herbicides, fungicides, and adjuvants in Red River Valley (RRV), Minnesota, are at significantly increased risk for certain types of birth defects including urogenital defects (Garry et al., 1996). Moreover, data showed a significant shift in the sex ratio of children borne with birth defects (Garry et al., 1996). Based on these epidemiological findings and health-reproductive survey data (unpublished), it was decided to explore whether certain commonly used herbicides, adjuvants, and fungicides affect specific developmental endpoints in vitro. At the cellular level, aneuploidy, alteration of programmed cell death (apoptosis), and altered cell proliferation due to endocrine disruption as well as direct cytotoxicity are known factors participating in teratogenesis and teratogenic events (Knudsen, 1990, 1997; Mittendorf, 1995; Colborn et al., 1993, 1998).

In these in vitro studies the MCF-7 assay (Soto et al., 1995) was adapted to flow cytometry to allow performance of semi-quantitative assessments of cytotoxicity, aneuploidy, cell proliferation, and apoptosis in the same cell system. Preliminary concentration range finding studies were carried out in a 72-h assay. This approach also allowed us to rapidly assess in vitro concentration-related cytotoxicity. Detailed cell proliferation studies were performed at the end of 7 d of culture. To further assess potential xenobiotic-mediated estrogenic versus nonestrogenic proliferative effects, cells were cultured separately in estrogen-deprived serum (charcoal-dextran treated) and in serum containing nominal amounts of estrogen (10^{-12} M) and other steroids.

In these initial studies, our screening efforts were focused based on (1) information gained from the earlier study of in vitro genotoxicity (Garry et al., 1999), (2) data from a reproductive health and pesticide use survey of the Red River Valley pesticide applicators and their families ($n = 835$, unpublished), and (3) data from the literature dealing with known or suspected developmental and reproductive toxicity. 2,4-D LV4 (2,4-dichlorophenoxy acetic acid, isooctyl ester) and 2,4-D amine (2,4-dichlorophenoxy acetic acid, dimethylamine salt) were selected on the basis of (1) common usage, (2) data from earlier in vitro genotoxicity study, and (3) negative reports regarding reproductive toxicity of the reagent-grade product (Garry et al., 1996). In the study design, data obtained from these commercial products were compared to results obtained from reagent-grade chemical. Roundup (glyphosate) was selected on the basis of common usage and earlier reported negative findings in regard to in vitro genotoxicity. Triphenyltin is the most commonly used fungicide in the Red River Valley. Depending on weather conditions, the fungicide may be applied by air or ground application. The usage of this fungicide in the RRV is also reported to be the second highest in the United States, only less than Georgia (Minnesota Star Tribune, 18 February 1998). Earlier studies by others suggest that triphenyltin may be a reproductive toxicant (WHO, 1980). Manzate (mancozeb) and Tilt (propiconazole) were selected based on their common usage in Red River Valley. The metabolite of mancozeb, ethylene thiourea

(ETU), has been reported to be carcinogenic and teratogenic in animal studies (WHO, 1988). ETU induces disruption of thyroid function and is causally related to thyroid cancer in animals (Steenland et al., 1997).

With regard to adjuvants selected, as a general rule, adjuvants are added to pesticide mixtures in order to aid operation and improve effectiveness of the active ingredients. These chemicals include wetting agents, spreaders, emulsifiers, dispersing agents, foaming adjuvants, foam suppressants, penetrants, etc. (Meister, 1997). The chemical composition of these mixtures can include polyoxyethylenes, polyvinyl compounds, and paraffin oils in the same commercial product (Foy, 1992; Harvey, 1992; Schonherr & Bauer, 1992). The toxicity, particularly developmental toxicity, of agricultural adjuvants and other "inert" ingredients in pesticide formulations has been and remains understudied. Given the available information regarding chemical composition and mode of action, it is possible that adjuvants may enhance toxicity of pesticides by increasing dermal penetration of the pesticide or may exert toxicity independent of the pesticide. The present study examines the latter hypothesis for four different adjuvant formulations commonly used in the RRV.

MATERIALS AND METHODS

Test Chemicals and Positive Controls

Commercial-grade pesticides and adjuvants used in these studies were obtained from pesticide applicators and the Minnesota Extension Service. Product composition and manufacturer were determined from the product label and/or package insert and available material safety data sheet (MSDS) information supplied by the manufacturer.

Herbicides were:

2,4-D LV4 (commercial grade, 66.2% 2,4-dichlorophenoxyacetic acid, isooctyl ester) (CENEX, St. Paul, MN).

2,4-D Amine (commercial grade, 46.5% 2,4-dichlorophenoxyacetic acid, dimethylamine salt) (Platte Chemical Company, Fremont, NE).

2,4-Dichlorophenoxyacetic acid, isooctyl ester (reagent grade) (Chem Service, West Chester, PA).

2,4-Dichlorophenoxyacetic acid (pure acid, reagent grade) (Sigma, St. Louis, MO).

Roundup (commercial grade, 0.99% glyphosate) (Monsanto, St. Louis, MO).

Glyphosate (reagent grade, *N*-phosphonomethyl-glycine, monoisopropylamine salt, 40 wt% solution in water) (Aldrich, Milwaukee, WI).

Fungicides were:

Triphenyltin acetate (reagent grade) (Chem Service, West Chester, PA).

Mancozeb (reagent grade) (Ultra Scientific, North Kingstown, RI).

Manzate (commercial grade, 75% mancozeb) (Dupont, Wilmington, DE).
 Tilt (reagent grade) (Ultra Scientific, North Kingstown, RI).
 Tilt (commercial grade, 42% propiconazole) (Ciba-Geigy, Tarrytown, NY).

Adjuvants were:

Activate Plus, a commercial surfactant mixture containing alkylaryl polyoxyethylene glycols and free fatty acids (Terra International Inc., Sioux City, IA).

Nalcotrol, a commercial-grade mixture containing polyvinyl polymer (polyarylamide) and inert ingredients (Nalco Chemical Company, Naperville, IL).

Preference, a commercial-grade mixture containing a proprietary surfactant containing less than 0.002% ethylene oxide and less than 0.002% dioxane (CENEX, St. Paul, MN).

X-77, a commercial surfactant mixture containing alkyl polyoxyethylenes, free fatty acids, glycol, and isopropanol (Loveland Industries, Greeley, CO).

Herbimax, a commercial mixture containing petroleum hydrocarbons and mono- and diesters of omega hydroxypolyoxyethylenealkyl polyoxyethylenes (Loveland Industries, Greeley, CO).

Positive controls were as follows:

1. For 72-h cytotoxicity screening assay: actinomycin D (Sigma, St. Louis, MO).
2. For 7-d estrogen-responsive proliferation assay: 17-beta-estradiol (Sigma, St. Louis, MO), diethylstilbestrol (DES) (Sigma, St. Louis, MO), and zearalenone (Sigma, St. Louis, MO).
3. For apoptosis and aneuploidy assay: paclitaxel (Taxol) (Sigma, St. Louis, MO).

Cell Line and Cell Culture

MCF-7 cells were kindly provided by Dr. B. Leung, originally from the Michigan Cancer Foundation, Detroit, MI. Cells were maintained in a humidified atmosphere containing 5% CO₂ in culture flasks containing phenol red-free DMEM/F-12 medium (GibcoBRL) supplemented with 10% heat-inactivated, non-charcoal-dextran-treated fetal bovine serum (Hyclone) and 1% gentamicin (GibcoBRL). Fetal bovine sera from same lot were used in all cultures. Cells were subcultured every 4 to 7 d and were tested for mycoplasma contamination (Mycotect kit-GibcoBRL). Passages 393–455 were used in the experiments.

72-Hour Cytotoxicity Screening Assay

MCF-7 cells were seeded at 10⁵ cells/ml in 24-well plates. Test chemicals, cytotoxicity positive control (actinomycin D), or solvent vehicle

was added within 2 to 4 h of incubation and cultured for 72 h in media plus 10% fetal bovine serum with no charcoal-dextran (CD) treatment. Flow cytometric measurements of cell number and cell viability were conducted. In brief, after harvest with 0.05% trypsin–ethylenediamine tetraacetic acid (EDTA) and resuspending in phosphate-buffered saline (PBS, pH 7.4), MCF-7 cells were stained with 10 µg/ml propidium iodide (PI) and analyzed in a FACSCalibur flow cytometer (Becton-Dickinson) equipped with a 488-nm argon laser, according to the method of Dittel and LeBien (1995). Viable cells exclude PI. Hence, cell viability was determined by percentage of cells not stained by PI (PI negative). All data were analyzed by Cell Quest software (Becton Dickinson). Concentration-related effects were noted and compared to controls for each study. At least three concentration points were examined in three separate studies for each study chemical. These screening data provided essential concentration range-finding studies and mechanistic information regarding cell death.

7-Day Proliferation Assay in MCF-7 Cells

With CD-Treated Serum Forty-eight hours prior to toxicity studies, MCF-7 cells were seeded in media plus 10% charcoal-dextran (CD) treated fetal bovine serum at 10^4 cells/ml in 24-well plates (Soto et al., 1994). Cells conditioned in media plus steroid growth-factor-deficient serum (CD) for 2 d were treated with different dilutions of test chemicals, 10^{-9} M estradiol (positive control), or solvent vehicle, added into wells in duplicate and mixed thoroughly. The final volume of solvent was less than 0.1% of the total volume in each well. After 7 d of culturing, cells were harvested and prepared for FACS (fluorescence-activated cell sorter) analysis. Cell number and viability were analyzed as described earlier. Increased cell numbers over control indicated estrogen-induced cell proliferation (Soto et al., 1994).

Without CD-Treated Serum Although MCF-7 cells proliferate in an estrogen-dependent manner, there are a number of other factors that may affect proliferation in this cell line. To address this question, fetal bovine serum in the culture media was not treated with charcoal-dextran (CD). Otherwise, conditions of culture and test chemical additions followed the same protocol already stated.

FACS Cell Cycle Analysis and Apoptosis

MCF-7 cells were seeded in T25 flasks at an initial density of 10^5 cells/ml in non-CD-treated serum. Twenty-four hours after seeding, the cells were incubated with triphenyltin (TPT), paclitaxel (Taxol), or vehicle at different concentration levels for the time indicated, then trypsinized, washed twice with PBS (pH 7.4), fixed with cold 70% ethanol, and stored at -20°C overnight. The cells were then treated with 50 µg/ml propidium iodide (PI) and 0.1 mg/ml ribonuclease A for 30 min, then analyzed with a

FACSCalibur flow cytometer (Darzynkiewicz et al., 1997). Doublets were excluded on FL2-W versus FL2-A plots (Becton Dickinson). DNA histograms were analyzed by ModFit LT software (Verity Software House, histograms not shown).

Cell Morphology and Apoptosis

Cells were seeded on glass coverslips in non-CD-treated serum in well cultures or glass chamber slides, treated with TPT, mancozeb, or vehicle 24 h after seeding, and incubated for an additional 24 h. The adherent cells were washed with PBS (pH 7.4), fixed with 70% ethanol at 4°C overnight, then stained with hematoxylin–eosin. The stained coverslips were mounted in Permount and examined by light microscopy. The fixed–stained preparations were examined qualitatively to identify apoptotic bodies, nuclear fragmentation, and nuclear chromosomal margination, to morphologically confirm the flow cytometric findings related to apoptosis (Allen et al., 1997; Darzynkiewicz et al., 1997).

Statistics

For each chemical, at least three separate studies were performed to establish concentration-response relationships. At least two replicates per concentration per experiment were used to establish each data point in the experiment. In cytotoxicity and cell proliferation assays, total cell counts from growth assays were normalized to control cell count in each experiment and cell numbers were compared as percent of control. Linear regression analysis was performed. The slope of the regression line was tested, if $\beta > 0$ (Daniel, 1987). Single-factor analysis of variance (ANOVA) was used to test non-linear data. In apoptosis assays, DNA histograms from at least three separate studies for each concentration and treatment time were fitted by ModFit Software (Verity Software House). Percentages of apoptotic cells in DNA histograms were calculated and tested by single-factor ANOVA and Scheffé's test of comparison of means (Gad & Weil, 1986; Daniel, 1987).

RESULTS

Adjuvants and Herbicides with Pseudo-Estrogen Activity

Table 1 summarizes the overall MCF-7 cell proliferative effects induced by treatment with positive control chemicals (17-beta-estradiol, diethylstilbestrol, or zearalenone), compared to treatment with adjuvants or herbicides listed. Concentrations tested for proliferation assay showed little or no cytotoxicity based on the 3-d cytotoxicity screening assay results (data not shown). As shown in the table, DES, not unexpectedly, is most active in induction of MCF-7 cell proliferation in terms of percent increase in cell number ($238 \pm 29.5\%$ of control) and concentration applied

(0.0027 µg/ml or 10^{-8} M) after 7 d of culture with charcoal-dextran-treated serum (standard conditions). Two of five adjuvants studied significantly increased cell number (X-77 and Activate Plus) at relatively low concentrations (0.01 and 0.1 µg/ml). The magnitude of the response (percent increase) is similar to that recorded for zearalenone, a fungal toxin and an agricultural environmental contaminant. The commercial-grade chlorophenoxy herbicides (2,4-D LV4 and 2,4-D amine) significantly induced MCF-7 cell proliferation but at a log unit higher concentration than the adjuvants. Comparison of the proliferative activity of commercial grade chlorophenoxy herbicides with reagent-grade herbicides (Figure 1) shows that reagent-grade chlorophenoxy products were inactive in the assay system. These data suggest that additives or contaminants in the commercial product are responsible for the estrogen-like receptor-mediated, proliferative activity.

In contrast, glyphosate and its commercial equivalent, Roundup, produced similar results in CD- (Table 1) and non-CD-treated cultures (not shown). The extent of maximum induction of cell proliferation is significant ($p < .05$, linear regression) with ($126 \pm 5.1\%$ and $135 \pm 3.5\%$) or without CD treatment ($121 \pm 10.3\%$ and $130 \pm 7.98\%$) for glyphosate

TABLE 1. Comparison of Potential In Vitro Estrogenicity: Test Chemicals Versus Positive Controls

Chemical	Concentration range that induces MCF-7 proliferation (µg/ml)	Maximum MCF-7 proliferation induced (% of control, mean $\pm$ SE)	Concentration induced maximum MCF-7 proliferation (µg/ml)
Positive control			
DES	0.000027–0.027 (10^{-12} M– 10^{-7} M)	238 ± 29.5^a	0.0027 (10^{-8} M)
17β-Estradiol	0.000027–0.027 (10^{-12} M– 10^{-7} M)	184 ± 13.1^a	0.0027 (10^{-8} M)
Zearalenone	0.00032–0.032 (10^{-11} M– 10^{-7} M)	157 ± 18.0^a	0.032 (10^{-7} M)
Commercial-grade herbicides			
2,4-D LV4	0.1–10	228 ± 30.3^a	1.0
2,4-D Amine	1–10	149 ± 17.8^a	1.0
Glyphosate	0.228–2.28 (10^{-5} M– 10^{-4} M)	135 ± 3.5^a	2.28 (10^{-4} M)
Roundup	1–10	126 ± 5.1^a	10
Adjuvants			
X-77	0.01–1	189 ± 7.2^a	0.1
Activate Plus	0.01–1	159 ± 11.6^a	0.1
Preference	0.01–0.1	143 ± 30.0	NR
Nacoltrol	NR	109 ± 9.2	NR
Herbimax	0.01–0.1	134 ± 14.4	NR

Note. All data were tested by linear regression. NR, nonresponsive.

^aSignificantly different from control ($p < .05$).

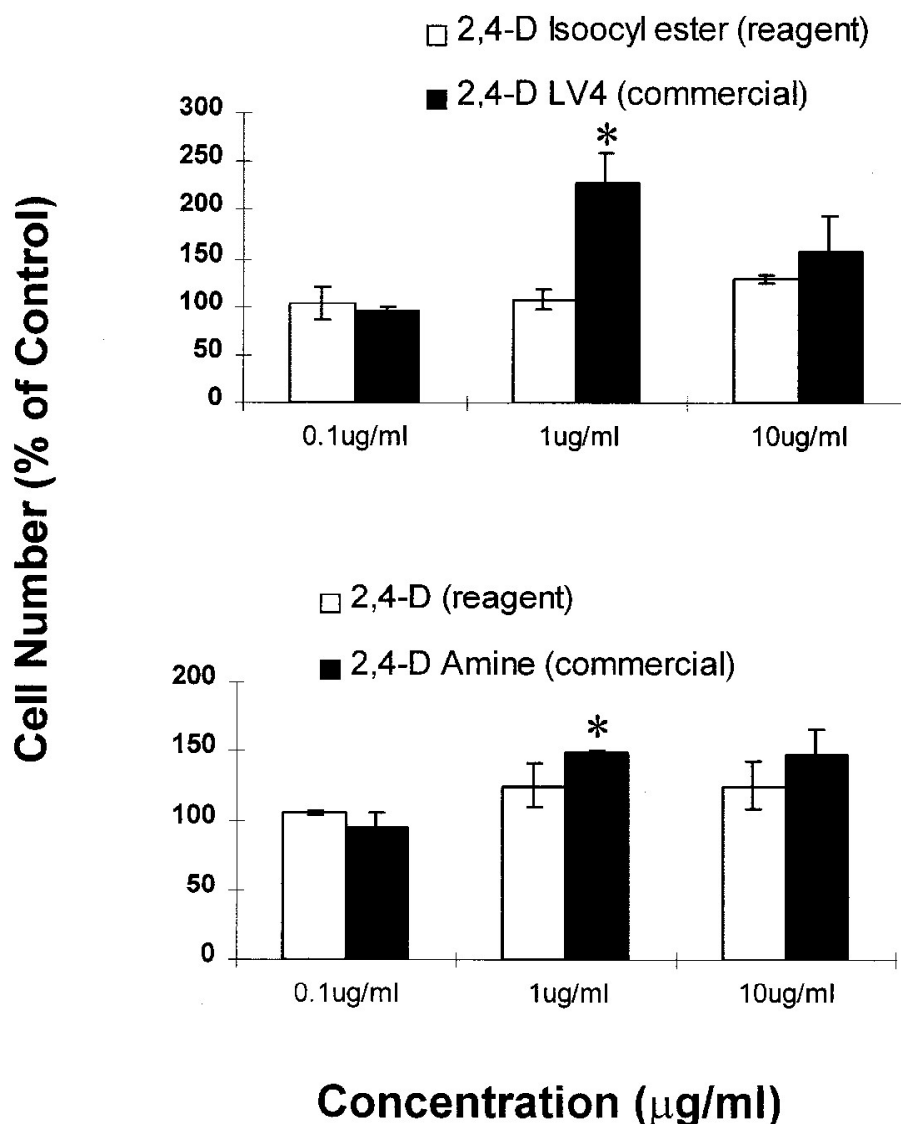


FIGURE 1. Effect of commercial herbicides 2,4-D LV4 and 2,4-D amine on proliferation in MCF-7 cells in the presence of charcoal-dextran (CD) treatment. Asterisk indicates significantly different from control ($p < .05$).

and Roundup, respectively; data suggest nonestrogenic induction of cell proliferation. Along these lines (Figure 2), DES induced MCF-7 cell proliferation in cultured MCF-7 cells with and without CD treatment, while 17-beta-estradiol induced significant cell proliferation only in CD-treated serum.

Apoptosis and Aneuploidy

In 3-d screening studies, all three fungicides (triphenyltin, mancozeb, and Tilt) induced cytotoxicity over the concentration range tested, with 50–70% decrease in cell number at the highest concentration tested (data not shown). However, only 2 (triphenyltin and mancozeb) of 16 pesticide products studied showed an apoptotic pattern of cell death by flow cytometry. All the other herbicides, adjuvants, and fungicides tested produced a necrotic pattern of cell death at test chemical concentrations that

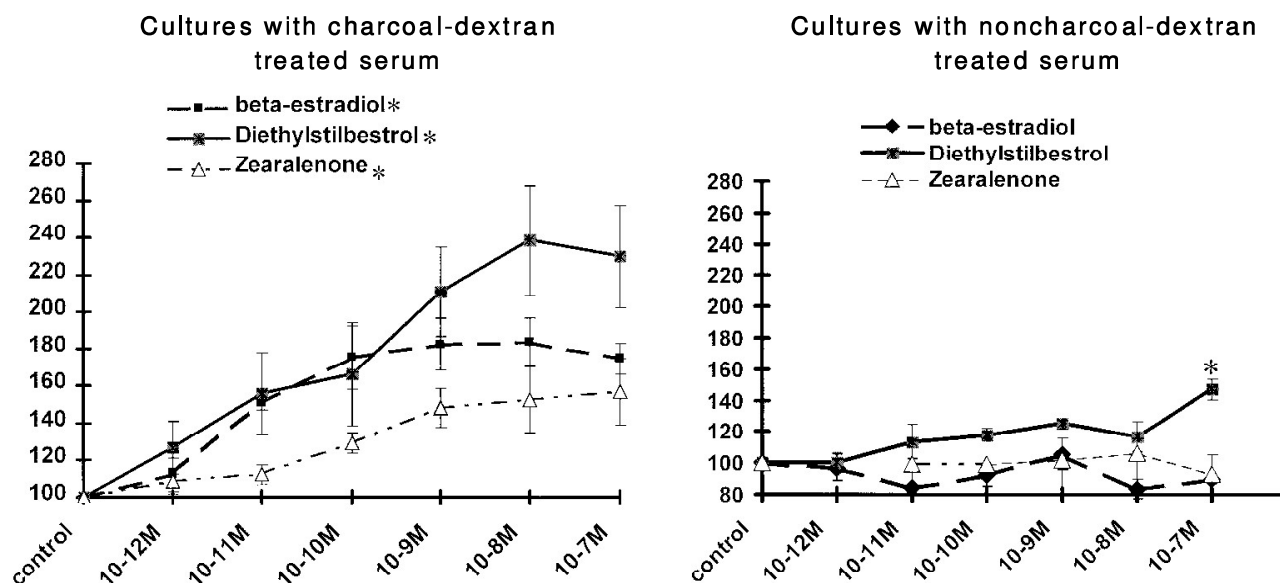


FIGURE 2. Effect of chemicals on MCF-7 proliferation in the absence or presence of charcoal-dextran (CD) treatment. Asterisk indicates significantly different from control ($p < .05$).

induced cytotoxicity, characterized by decreased cell viability and loss of cell membrane integrity (data not shown). As shown in Figure 3, progressive DNA degradation and formation of a sub-G1 DNA peak occurred between 24 and 48 h at 4.1 $\mu\text{g}/\text{ml}$ (10^{-5} M), a lethal concentration of triphenyltin in MCF-7 cells. Mancozeb demonstrated a similar cell death pattern at 10 and 50 $\mu\text{g}/\text{ml}$ (0.38×10^{-4} M and 1.88×10^{-4} M) in the same time frame. The metabolite of mancozeb, ethylenethiourea (ETU), pro-

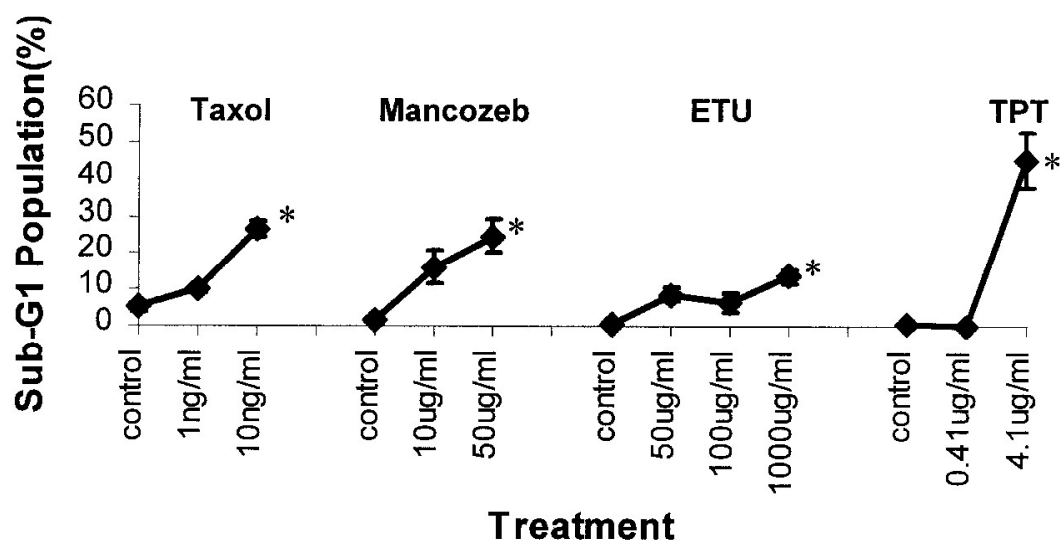


FIGURE 3. The fungicide triphenyltin (TPT), mancozeb, and its metabolite ethylenethiourea (ETU) induced apoptosis. MCF-7 cells treated by TPT, mancozeb, and ETU developed a "sub-G1" peak (formation of free apoptotic bodies) (Darzynkiewicz et al., 1997) in FACS-generated DNA histograms (not shown). The concentration range of the test chemicals that induced sub-G1 peaks was compared with that of Taxol, the positive control for apoptosis. Treatment times that induced maximum percentage sub-G1 population with no massive DNA degradation were compared among test chemicals and Taxol (TPT, mancozeb, and ETU, 24 h; Taxol, 48 h). Asterisk indicates significantly different from control ($p < .05$, ANOVA).

duced apoptotic induction at a much higher concentration (1000 $\mu\text{g/ml}$ or 0.01 M). Microscopic morphologic examination of MCF-7 cells treated with these fungicides demonstrated the presence of apoptotic bodies and condensed chromatin (photos not shown; Wyllie, 1980; Darzynkiewicz et al., 1997; Knudsen, 1997; Allen et al., 1997). For TPT, at 0.41 $\mu\text{g/ml}$ (10^{-6} M), a sublethal concentration (Figure 4), G2/M arrest was noted, indicating an aneugenic effect. Neither mancozeb nor its metabolite (ETU) induced an aneugenic effect. The third fungicide tested, Tilt (propiconazole), produced necrosis in MCF-7 cells at 10–100 $\mu\text{g/ml}$, characterized by loss of cell membrane integrity and decreased cell number and viability in 72-h screening assay (data not shown).

DISCUSSION

Environmental Estrogenic Effects Detected

In recent years, concerns have been raised about the participation of chemicals with endocrine activity in birth defects and other forms of

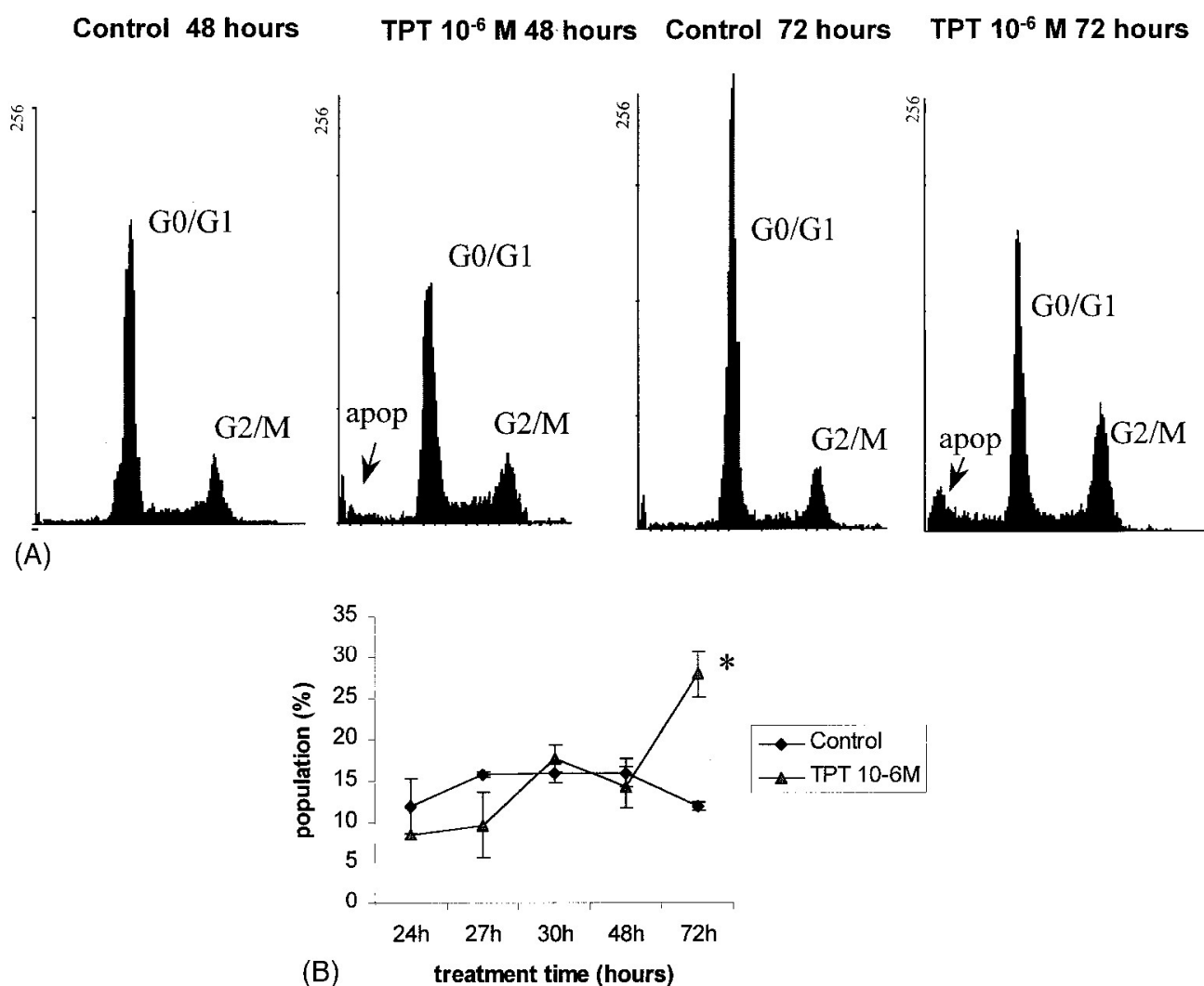


FIGURE 4. FACS-generated DNA histograms of TPT-induced aneuploidy in MCF-7 cells. Asterisk indicates significantly different from control ($p < .05$). (A) FACS analysis. (B) Time course.

reproductive and developmental toxicity (Colborn et al., 1993; Korach & McLachlan, 1995; Kavlock et al., 1996). More than 50 pesticides, industrial chemicals, and their by-products have been identified as environmental estrogens, which differ widely in structure, point of origin, degree and extent of human exposure, and persistence (Fairly, 1996; Kavlock et al., 1996; Zacharewski, 1998). Aside from the known overt developmental effects of dioxins and polychlorinated biphenols (PCBs) in humans at relatively low concentrations (Safe et al., 1991), there is little evidence available to indicate human developmental toxicity for major classes of industrial chemicals including bisphenol A and other alkylphenols (Sharpe et al., 1995; Nimrod & Benson, 1996). Some therapeutic agents, such as diethylstilbestrol (DES) and ethynyl estradiol, are well-documented reproductive toxins active in relatively low concentrations (Mittendorf, 1995). Dietary phytoestrogens, such as genistein, are suspected human reproductive toxins but lack well-documented evidence for adverse reproductive effects (Dees et al., 1997). Environmental phytoestrogens such as zearalenone, a fungal toxin and contaminant of wheat, are noted reproductive toxins in animals (Mayr, 1988). From this perspective, the concentration-related changes in MCF-7 cell proliferation seen in these *in vitro* studies of adjuvants and the commercial-grade chlorophenoxy herbicides may not be biologically significant. Concentration levels inducing cell proliferation in these studies suggest that the effects of DES, a proven biologically active pseudoestrogen, are at least three orders of magnitude lower than for any of the adjuvants and commercial-grade chlorophenoxy herbicides studied. In the context of the present work, cumulative effects cannot be considered.

The present study demonstrates that flow cytometry measurement of cell number, viability, and apoptosis can be a useful screening and detection method. Moreover, use of non-CD treated sera shows some promise as a means of detection of nonestrogenic chemically induced proliferation in MCF-7 cells (Figure 2). Other single-parameter cell assay systems (Soto et al., 1995), while effective for the endpoints studied, do not allow for more broad-spectrum developmental analysis as demonstrated here.

With regard to specific estrogen-like effects, the adjuvants X-77 and Activate Plus induced MCF-7 proliferation in CD-treated serum. These adjuvants are commercial-grade mixtures of alkylphenols (APs) and alkylphenol ethoxylates (APEs). Lower molecular weight nonyl- and octylphenol ethoxylates are used as contraceptives and spermicides and have been associated with other reproductive and developmental toxicity in animal studies (Nimrod & Benson, 1996). It has also been suggested that APs and APEs can be biotransformed into short-chain derivatives, such as nonyl- and octylphenols, and exert estrogenicity (Sharpe et al., 1995). Whether biotransformation *in vitro* and/or contamination of the commercial product with short-chain alkylphenol derivatives could be factors is unclear. Two adjuvants did not induce proliferation: Nalcotrol is not known to contain alkylphenols, and Preference contains polyoxyethylenes but not alkylphenols. Commer-

cial-grade 2,4-D products, but not the reagent-grade active ingredients, induced MCF-7 cell proliferation. Both 2,4-D commercial products contain 40–50% “inert ingredients.” Whether adjuvants are included in these commercial formulations as inert ingredients is unknown.

Toxicants, Non-CD-Treated Sera, and MCF-7 Cell Proliferation

Diethylstilbestrol (DES), the herbicide Roundup, and its active ingredient glyphosate induced MCF-7 cell proliferation with and without CD treatment. None of the other herbicides or adjuvants studied nor 17-beta-estradiol augmented cell proliferation when MCF-7 cells when grown in non-CD-treated serum.

CD (charcoal-dextran)-stripped and phenol-red-free serum has been widely used in in vitro estrogenicity assays to extract and thereby eliminate endogenous estrogens (Katzenellenbogen et al., 1987; Soto et al., 1994). However, charcoal-dextran extraction can also strip serum of other steroid hormones, insulin, growth hormone, thyroxine and glucagon (Cresto et al., 1972; Leclercq-Meyer et al., 1975; Pazzagli et al., 1975). These known hormonal modifiers of cell growth and other less well defined growth factors can arguably modify MCF-7 cell growth independent of estrogens (de Launoit & Kiss, 1989; Tada et al., 1993; Hafner et al., 1996; Nogueira & Brentani, 1996; Kudoh et al., 1996). In support of this contention, the addition of 10^{-12} M beta-estradiol alone (the concentration in whole-serum-enriched culture medium) in CD-treated cultures did not fully restore cell number to the non-CD-treated baseline growth level, indicating that other serum factors contribute to MCF-7 cell proliferation. Further, xenobiotic-induced proliferative activity in fully constituted serum (non-CD-treated) may indicate nonestrogenic induced cell proliferation. In support of this notion, addition of 17-beta-estradiol to cells in media containing fully constituted serum over concentrations ranging from 10^{-12} to 10^{-7} M did not augment cell proliferation in non-CD-treated serum but did induce concentration-dependent increased proliferation in media containing CD-treated serum. Thus, it appears that fully constituted serum masks or attenuates estrogen-mediated proliferative effects, whereas DES and glyphosate expression are relatively unaffected. Aside from DES actions as a pseudo-estrogenic developmental toxin, these data suggest that DES can have other nonestrogenic developmental effects. In line with this hypothesis, recent in vivo studies by Miller et al. (1998) demonstrate that DES disrupts *hox* gene expression during development.

In another vein, one of the nagging questions regarding in vitro environmental estrogen evaluation and detection of weak estrogenlike activity is whether these chemicals can have any clinically significant effects in vivo. Cultured cells deprived of background estrogens and nonestrogenic growth factors (CD-treated serum) may be sensitized to estrogenic stimuli (Masamura et al., 1995; Shim et al., 2000). Moreover, the in vivo response to exogenous estrogenic stimuli in a hormonally balanced milieu

may well be dampened. Based on this assumption, we speculate that acute exposure to high levels of weak pseudoestrogens would not produce clinically significant changes in the level of sex hormones in exposed persons. Consistent with this hypothesis, in current published work we and others recorded minor though statistically significant changes in sex hormone levels from male workers acutely exposed to pesticides and or herbicides containing pseudoestrogens (Garry et al., 1999; Straube et al., 1999). Whether pseudoestrogen-induced minor changes in sex hormone levels in women have biologic significance is poorly understood at the moment.

Apoptosis and Aneuploidy

Disturbance of apoptosis (programmed cell death) may alter the dynamic balance between cell elimination and proliferation in normal development (Knudsen, 1990, 1997). Triphenyltin (TPT) is the most commonly used fungicide in Red River Valley. TPT, although not a teratogen to mice and rats, induces fetal death and postimplantation embryonic mortality in these species (WHO, 1980; Noda et al., 1991; Ema et al., 1997, 1999). Previously, TPT was reported to induce apoptosis in several cell lines including HL-60 human leukemia cells and PC-12 pheochromocytoma-derived cells (Viviani et al., 1995; Marinovich et al., 1996). The present study confirms these earlier findings and notes aneugenic effects as well. Aneuploidy and polyploidy reflect delayed or halted cell cycle progression, which may interrupt normal cell proliferation or trigger improper apoptosis in developing embryos, and is linked to stillbirths and birth defects (Oshimura & Barrett, 1986; Schuler et al., 1998). TPT was reported to induce aneuploidy in cultured human lymphocytes by altering spindle function (Jensen et al., 1991). In the present study, TPT produced increased G2/M population under treatment with a sublethal concentration. By contrast, no aneuploidy was found with commercial- and reagent-grade mancozeb and its carcinogenic metabolite.

Manzate (mancozeb) and its metabolite, ethylenethiourea (ETU), have been reported to be endocrine disruptive, carcinogenic, and teratogenic (WHO, 1988; Tsuchiya et al., 1991; Colborn et al., 1993; Hurley et al., 1998). Occupational exposure to Manzate was also associated with increased chromosome aberrations and sister chromatid exchanges (Steenland et al., 1997). However, to our knowledge, there are no mechanistic reports regarding cell death patterns relevant to developmental effects of this widely used fungicide. The apoptotic effects found in our test system offer some insight into the mechanism of action of these fungicides.

The third fungicide tested, Tilt (propiconazole), produced necrosis in MCF-7 cells. This chemical has been reported to be an inhibitor of fungal toxin biosynthesis (ergosterol) and a P-450 liver enzyme inducer in rodents (Vanden Bossche et al., 1989; Cabral et al., 1991; Ronis et al., 1994). Embryotoxicity and teratogenicity of some triazole antifungal agents were

noted in animals and thought to be related to *Hox* gene expression (Flint & Boyle, 1985; Brown et al., 1992). Ketoconazole, an antifungal drug structurally related to propiconazole, has been reported to induce apoptosis through p53-dependent pathways in human hepatocarcinoma and colorectal carcinoma cell lines (Ho et al., 1998). Propiconazole was found to induce skeletal and visceral malformations in rats (Hassan, 1993). In this study, it was shown that necrosis, the predominant mechanism of cell death, occurred with both commercial- and reagent-grade propiconazole. More detailed assessments are underway to further study these findings in order to provide information about these fungicides' potential developmental toxicity.

Lastly, comparison of the FACS data for taxol and TPT in our studies suggests a somewhat similar mode of action. Paclitaxel (Taxol), the positive control in the apoptosis and aneuploidy studies, is known to induce apoptosis in MCF-7 cells (Saunders et al., 1997) and murine mammary and ovary tumors (Milas et al., 1995). Taxol was found to cause cyclin B1 accumulation and cyclin B1-dependent kinase activation, indicating G2/M arrest (Ling et al., 1998). Such mitotic failure may play a role in triggering apoptosis (Sorger et al., 1997; Ling et al., 1998). More detailed assessments are underway to further study the mechanistic relationship between aneuploidy and apoptosis induced by TPT, and implications for developmental toxicity (Ford, 1997; Aardema et al., 1998).

REFERENCES

- Aardema, M. J., Albertini, S., Arni, P., Henderson, L. M., Kirsch-Volders, M., Mackay, J. M., Sarraf, A. M., Stringer, D. A., and Taalman, R. D. 1998. Aneuploidy: A report of an ECETOC task force. *Mutat. Res.* 410:3–79.
- Allen, R. T., Hunter, W. J. 3rd, and Agrawal, D. K. 1997. Morphological and biochemical characterization and analysis of apoptosis. *J. Pharmacol. Toxicol. Methods* 37: 215–228.
- Brown, N. A., Hunt, P., and Krumlauf, R. 1992. Craniofacial developmental abnormalities induced by triazoles: a comparison of Homeobox gene expression and stage-specificity with retinoid acid-induced defects. *Teratology* 45:456.
- Cabral, R., Hoshiya, T., Hakoi, K., Hasegawa, R., Fukushima, S., and Ito, N. 1991. A rapid in vivo bioassay for the carcinogenicity of pesticides. *Tumori* 77:185–188.
- Colborn, T., vom Saal, F. S., and Soto, A. M. 1993. Developmental effects of endocrine-disrupting chemicals in wildlife and humans. *Environ. Health Perspect.* 101:378–384.
- Colborn, T., Smolen, M. J., and Rolland, R. 1998. Environmental neurotoxic effects: the search for new protocols in functional teratology. *Toxicol. Ind. Health* 14(1–2):9–23.
- Cresto, J. C., Dujovne, I. L., Castellani, P. I., Mitta, E. A., De Majo, S. F., and Foglia, V. G. 1972. Insulin radioimmunoassay by the charcoal-dextran technique. *Diabetologia* 8:292–295.
- Czeizel, A. E., Elek, C., Gundy, S., Metneki, J., Nemes, E., Reis, A., Sperling, K., Timar, L., Tusnady, G., and Viragh, Z. 1993. Environmental trichlorfon and cluster of congenital abnormalities. *Lancet* 341:539–542.
- Daniel, W. W. 1987. *Biostatistics: A foundation for analysis in health sciences*, 4th ed., pp. 367–415. New York: John Wiley and Sons.
- Darzynkiewicz, Z., Juan, G., Li, X., Gorczyca, W., Murakami, T., and Traganos, F. 1997. Cytometry in cell necrobiology: Analysis of apoptosis and accidental cell death (necrosis). *Cytometry* 27:1–20.

- Dees, C., Foster, J. S., Ahamed, S., and Wimalasena, J. 1997. Dietary estrogens stimulate human breast cells to enter the cell cycle. *Environ. Health Perspect.* 105(suppl. 3):633–636.
- de Launoit, Y., and Kiss, R. 1989. Influence of L-thyroxine, L-triiodothyronine, thyroid stimulating hormone, or estradiol on the cell kinetics of cultured mammary cancer cells. *In Vitro Cell. Dev. Biol.* 25:585–591.
- Dittel, B. N., and LeBien, T. W. 1995. The growth response to IL-7 during normal human B cell ontogeny is restricted to B-lineage cells expressing CD34. *J. Immunol.* 154:58–67.
- Ema, M., Miyawaki, E., Harazono, A., and Ogawa, Y. 1997. Effects of triphenyltin chloride on implantation and pregnancy in rats. *Reprod. Toxicol.* 11:201–206.
- Ema, M., Miyawaki, K., and Kawashima, K. 1999. Developmental toxicity of triphenyltin chloride after administration on three consecutive days during organogenesis in rats. *Bull. Environ. Contam. Toxicol.* 62:363–370.
- Fairly, P. 1996. Endocrine disruptors: Sensational or science? *Chem. Week.* 158:29–35.
- Flint, O. P., and Boyle, F. T. 1985. An in vitro test for teratogens: its application in the selection of non-teratogenic triazole antifungals. *Concepts Toxicol.* 3:29–35.
- Ford, J. H. 1997. Does aneuploidy per se cause developmental abnormalities? *Mutat. Res.* 396:195–203.
- Foy, C. L., ed. 1992. *Adjuvants for agrochemicals*. Boston: CRC Press.
- Gad, S., and Weil, C. S. 1986. *Statistics and experimental design for toxicologists*, pp. 94–98. Caldwell, NJ: Telford Press.
- Garry, V. F., Schreinemachers, D., Harkins, M. E., and Griffith, J. 1996. Pesticide applicers, biocides, and birth defects in rural Minnesota. *Environ. Health Perspect.* 104:394–399.
- Garry, V. F., Burroughs, B., Tarone, R., and Kesner, J. S. 1999. Herbicides and adjuvants: An evolving view. *Toxicol. Ind. Health* 15:159–167.
- Hafner, F., Holler, E., and von Angerer, E. 1996. Effect of growth factors on estrogen receptor mediated gene expression. *J. Steroid Biochem. Mol. Biol.* 58:385–393.
- Harvey, L. T., ed. 1992. *A guide to agricultural spray adjuvants used in the United States*. Fresno, CA: Thompson.
- Hassan, M. S. 1993. Embryotoxic and teratogenic effects of the organic fungicide Tilt in albino rats. *Bull. Faculty Pharm.* 31:459–463.
- Ho, Y., Tsai, P., Yu, C., Liu, H., Chen, R., and Lin, J. 1998. Ketoconazole-induced apoptosis through p53-dependent pathway in human colorectal and hepatocellular carcinoma cell lines. *Teratology* 153:39–47.
- Hurley, P., Hill, R., and Whiting, R. 1998. Mode of carcinogenic action of pesticides inducing thyroid follicular cell tumors in rodents. *Environ. Health Perspect.* 106:437–445.
- Jensen, K. G., Andersen, O., and Ronne, M. 1991. Organotin compounds induce aneuploidy in human peripheral lymphocytes in vitro. *Mutat. Res.* 246:109–112.
- Katzenellenbogen, B. S., Kendra, K. L., Norman, M. J., and Berthois, Y. 1987. Proliferation, hormonal responsiveness, and estrogen receptor content of MCF-7 human breast cancer cells grown in the short-term and long-term absence of estrogens. *Cancer Res.* 47:4355–4360.
- Kavlock, R. J., Daston, G. P., DeRosa, C., Fenner-Crisp, P., Gray, L. E., Kaattari, S., Lucier, G., Luster, M., Mac, M. J., Maczka, C., Miller, R., Moore, J., Rolland, R., Scott, G., Sheehan, D. M., Sinks, T., and Tilson, H. A. 1996. Research needs for the risk assessment of health and environmental effects of endocrine disruptors: A report of the U.S. EPA-sponsored workshop. *Environ. Health Perspect.* 104(suppl. 4):715–740.
- Knudsen, T. B. 1990. In vitro approaches to the study of embryonic cell death. In *In vitro methods in developmental toxicology*, eds. G. L. Kimmel and D. M. Kochhar, pp. 130–131. Boca Raton, FL: CRC Press.
- Knudsen, T. B. 1997. Cellular techniques for teratological cell death. In *Methods in toxicology*, ed. E. J. Massaro, pp. 184–190. Boca Raton, FL: CRC Press.
- Korach, K. S., and McLachlan, J. A. 1995. Techniques for detection of estrogenicity. *Environ. Health Perspect.* 103(suppl.)7:5–8.
- Kristensen, P., Irgens, L. M., Andersen, A., Bye, A. S., and Sundheim, L. 1997. Birth defects among offspring of Norwegian farmers, 1967–1991. *Epidemiology* 8:537–544.

- Kudoh, M., Susaki, Y., Ideyama, Y., Nanya, T., Mori, M., Shikama, H., and Fujikura, T. 1996. Inhibitory effect of a novel non-steroidal aromatase inhibitor, YM511 on the proliferation of MCF-7 human breast cancer cell. *J. Steroid Biochem. Mol. Biol.* 58:189–194.
- Leclercq-Meyer, V., Marchand, J., Rebolledo, O., Malaisse, W. J., and Leclercq, R. 1975. A combined radioimmunoassay for glucagon and insulin. *Diabetologia* 11:419–425.
- Ling, Y. H., Consoli, U., Tornos, C., Andreeff, M., and Perez-Soler, R. 1998. Accumulation of cyclin B1, activation of cyclin B1-dependent kinase and induction of programmed cell death in human epidermoid carcinoma KB cells treated with taxol. *Int. J. Cancer* 75:925–932.
- Marinovitch, M., Viviani, B., Corsini, E., Ghilardi, F., and Galli, C. L. 1996. NF-kappaB activation by triphenyltin triggers apoptosis in HL-60 cells. *Exp. Cell Res.* 226:98–104.
- Masamura, S., Santner, S. J., Heitjan, D. F., and Santen, R. J. 1995. Estrogen deprivation causes estradiol hypersensitivity in human breast cancer cells. *J. Clin. Endocrinol. Metab.* 80:2918–2925.
- Mattison, D. R., Bogumil R. J., Chapin, R., Hatch, M., Hendricks, A., Jarrell, J., LaBarbera A. L., Schrader, S. M., and Selevan, S. 1990. Reproductive effects of pesticides. In *Effects of pesticides on human health* (Advances in Modern Environmental Toxicology, vol. 18), eds. R. S. Baker, C. F. Wilkinson, pp. 297–348. Princeton, NJ: Princeton Scientific.
- Mayr, U. E. 1988. Estrogen-controlled gene expression in tissue culture cells by zearalenone. *FEBS Lett.* 239:223–226.
- Meister, R. 1997. *Farm chemicals handbook '97*, p. C10. Willoghby, OH: Meister Publishing.
- Milas, L., Hunter, N. R., Kurdoglu, B., Mason, K. A., Meyn, R. E., Stephens, L. C., and Peters, L. J. 1995. Kinetics of mitotic arrest and apoptosis in murine mammary and ovarian tumors treated with taxol. *Cancer Chemother. Pharmacol.* 35:297–303.
- Miller, C., Degenhardt, K., and Sassoon, D. A. 1998. Fetal exposure to DES results in deregulation of Wnt7a during uterine morphogenesis. *Nature Genet.* 20:228–230.
- Mittendorf, R. 1995. Teratogen update: Carcinogenesis and teratogenesis associated with exposure to diethylstilbestrol (DES) in utero. *Teratology* 51:435–445.
- Murakami, M., and Fukami, J. 1986. Relationship between specific molecular connectivity indices and teratogenicity, carcinogenicity, and mutagenicity of carbamate pesticides. *Bull. Environ. Contam. Toxicol.* 37:326–329.
- Nimrod, A. C., and Benson, W. H. 1996. Environmental estrogenic effects of alkylphenol ethoxylates. *Crit. Rev. Toxicol.* 26: 335–364.
- Noda, T., Morita, S., Yamano, T., Shimizu, M., and Yamada, A. 1991. Effects of triphenyltin acetate on pregnancy in rats by oral administration. *Toxicol. Lett.* 56:207–212.
- Nogueira, C. R., and Brentani, M. M. 1996. Triiodothyronine mimics the effects of estrogen in breast cancer cell lines. *J. Steroid Biochem. Mol. Biol.* 59:271–279.
- Oshimura, M., and Barrett, J. C. 1986. Chemically induced aneuploidy in mammalian cells: Mechanisms and biological significance in cancer. *Environ. Carcinogen.* 8:129–159.
- Pazzagli, M., Forti, G., Cappellini, A., and Serio, M. 1975. Radioimmunoassay of plasma dihydrotestosterone in normal and hypogonadal men. *Clin. Endocrinol.* 4:513–520.
- Poynor, D. H., Lupkiewicz, S., Sage, S. R., Carver, V. H., Kousseff, B. G., Lubs, H. A., and Williams, C. A. 1997. Paternal exposures and the question of birth defects. *J. Florida Med. Assoc.* 84:323–326.
- Roeleveld, N., Zielhuis, G. A., and Gabreels, F. 1990. Occupational exposure and defects of the central nervous system in offspring: Review. *Br. J. Ind. Med.* 47:580–588.
- Ronis, M. J., Ingelman-Sundberg, M., and Badger, T. M. 1994. Induction, suppression and inhibition of multiple hepatic cytochrome P450 isozymes in the male rat and bobwhite quail (*Colinus virginianus*) by ergosterol biosynthesis inhibiting fungicides (EBIFs). *Biochem. Pharmacol.* 48:1953–1965.
- Safe, S., Astroff, B., Harris, M., Zacharewski, T., Dickerson, R., Romkes, M., and Biegel, L. 1991. 2,3,7,8-Tetrachlorodibenzo-*p*-dioxin (TCDD) and related compounds as antioestrogens: Characterization and mechanism of action. *Pharmacol. Toxicol.* 69:400–409.
- Saunders, D. E., Lawrence, W. D., Christensen, C., Wappler, N. L., Ruan, H., and Deppe, G. 1997. Paclitaxel-induced apoptosis in MCF-7 breast-cancer cells. *Int. J. Cancer* 70:214–220.

- Schonherr, J., and Bauer, H., eds. 1992. Analysis of effects on surfactants on permeability of plant cultures. In *Adjuvants for agrichemicals*, pp. 17–35. Boca Raton, FL: CRC Press.
- Schuler, M., Hasegawa, L., Parks, R., Metzler, M., and Eastmond, D. A. 1998. Dose-response studies of the induction of hyperdiploidy and polyploidy by diethylstilbestrol and 17 beta-estradiol in cultured human lymphocytes using multicolor fluorescence in situ hybridization. *Environ. Mol. Mutagen.* 31:263–273.
- Sharpe, R. M., Fisher, J. S., Millar, M. M., Jobling, S., and Sumpter, J. P. 1995. Gestational and lactational exposure of rats to xenoestrogens results in reduced testicular size and sperm production. *Environ. Health Perspect.* 103:1136–1143.
- Shim, W. S., Conaway, M., Masamura, S., Yue, W., Wang, J. P., Kumar, R., and Santen, R. J. 2000. Estradiol hypersensitivity and mitogen-activated protein kinase expression in long-term estrogen deprived human breast cancer cells in vivo. *Endocrinology* 141:396–405.
- Sorger, P. K., Dobles, M., Tournabize, R., and Hyman, A. A. 1997. Coupling cell division and cell death to microtubule dynamic. *Curr. Opin. Cell Biol.* 9:807–814.
- Soto, A. M., Chung, K. L., and Sonnenschein, C. 1994. The pesticides endosulfan, toxaphene, and dieldrin have estrogenic effects on human estrogen-sensitive cells. *Environ. Health Perspect.* 102:380–383.
- Soto, A. M., Sonnenschein, C., Chung, K. L., Fernandez, M. F., Olea, N., and Serrano, F. O. 1995. The E-SCREEN assay as a tool to identify estrogens: an update on estrogenic environmental pollutants. *Environ. Health Perspect.* 103(suppl. 7):113–122.
- Steenland, K., Cedillo, L., Tucker, J., Hines, C., Sorensen, K., Deddens, J., and Cruz, V. 1997. Thyroid hormones and cytogenetic outcomes in backpack sprayers using ethylenebis(dithiocarbamate) (EBDC) fungicides in Mexico. *Environ. Health Perspect.* 105:1126–1130.
- Straube, E., Straube, W., Kruger, E., Bradatsch, M., Jacob-Meisel, M., and Rose, H. J. 1999. Disruption of male sex hormones with regard to pesticides: Pathophysiological and regulatory aspects. *Toxicol. Lett.* 107:225–231.
- Tada, A., Sasaki, H., Nakamura, J., Yoshihama, M., and Terashima, Y. 1993. Aromatase activity and the effect of estradiol and testosterone on DNA synthesis in endometrial carcinoma cell lines. *J. Steroid Biochem. Mol. Biol.* 44:661–666.
- Tsuchiya, T., Takanashi, A., Asada, S., Takakubo, F., Ohsumi-Yamashita, N., and Eto, K. 1991. Comparative studies of embryotoxic action of ethylenethiourea in rat whole embryo and embryo cell culture. *Teratology* 43:319–324.
- Vanden Bossche, H., Marichal, P., Gorrens, J., Coene, M. C., Willemsens, G., Bellens, D., Roels, I., Moereels, H., and Janssen, P. A. 1989. Biochemical approaches to selective antifungal activity. Focus on azole antifungals. *Mycoses* 32(suppl. 1):35–52.
- Viviani, B., Rossi, A. D., Chow, S. C., and Nicotera, P. 1995. Organotin compounds induce calcium overload and apoptosis in PC12 cells. *Neurotoxicology* 16:19–26.
- Wang, G. M. 1988. Regulatory decision making and the need for and the use of exposure data on pesticides determined to be teratogenic in test animals. *Teratogen. Carcinogen. Mutagen.* 8:117–126.
- World Health Organization. 1980. *Environmental health criteria 15. Tin and organotin compounds: A preliminary review*, pp. 63–92. Geneva: WHO.
- World Health Organization. 1988. *Environmental health criteria 78. Dithiocarbamate pesticides, ethylenethiourea, and propylenethiourea: A general introduction*, pp. 92–101. Geneva: WHO.
- Wyllie, A. 1980. Cell death: The significance of apoptosis. *Int. Rev. Cytol.* 68:251–306.
- Zacharewski, T. 1998. Identification and assessment of endocrine disruptors: Limitations of in vivo and in vitro assays. *Environ. Health Perspect.* 106(suppl. 2):577–582.